

Fundamentals of Inference

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Module 10 Overview

- More on Bayesian modeling
- Examples using the R Software
- Calibrated Bayes for Checking
- Superpopulation Modeling

Some Standard Univariate Models

- Setup:

$$\textit{Population} : Y = \{Y_1, Y_2, \dots, Y_N\}$$

$$\textit{Sample} : y_{inc} = \{y_1, y_2, \dots, y_n\} \subset Y$$

$$\textit{Design} : \Pr(I_j = 1) = \pi_j, j = 1, 2, \dots, N$$

$$\textit{Model} : \Pr(Y) = \Pr(Y_1, Y_2, \dots, Y_N)$$

$$Y_{exc} = Y \cap y_{inc}^c$$

$$\textit{Inference} : \Pr(Y_{exc} \mid y_{inc})$$

- **Normal Continuous Variables:**

$$Y_i \mid \mu, \sigma^2 : iid N(\mu, \sigma^2)$$

Prior Distribution: $\pi(\mu, \sigma)$

- Bayesian Inference:

- Simulate draws (posterior, then excluded cases):

$$(\mu, \sigma) \mid y_{inc}$$

$$Y_{exc} \mid \mu, \sigma$$

- Create # of R potential populations, compute estimands of interest for each, then combine

$$Y^{(r)} = (y_{inc}, Y_{exc}^{(r)}), r = 1, 2, \dots, R$$

$$Q^{(r)} = Q(Y^{(r)})$$

- **Non-Normal Continuous Variables:**
 - Consider transformation to normality
 - For example, income, $Y^{1/3}$ *or* $\log Y$
 - Generate draws on the transformed scale and transform back to the original scale
- Box-Cox Transformations:
$$g_{\lambda}(y) = \frac{Y^{\lambda} - 1}{\lambda}$$
- Consider Non-Normal continuous distributions
 - Gamma, Weibull, t, chi-square, etc.

- **Count Variables:**

$$Y_i \mid \lambda : iid \text{ Poisson}(\lambda)$$

Prior: $\pi(\lambda)$

$$Y_i \mid \lambda : iid \text{ Negative} - \text{Binomial}(\lambda)$$

Prior: $\pi(\lambda)$

- **Polytomous Variables:**

- Multinomial model
- Ordered categorical model

Justification of Design-Based Confidence Intervals as Bayesian Credible intervals

- Consider the Simple Random Sample case...
- It is possible to identify priors such that the resulting credible interval will be equal to a confidence interval (although interpretations are different)

- Binomial Example: $Y | \theta : \text{Ber}(1, \theta)$

$$\pi(\theta) \propto \theta^{-1} (1 - \theta)^{-1}$$

$$\text{Improper Prior: } \int_0^1 \pi(\theta) d\theta = \infty$$

- Normal Example: $Y | \mu, \sigma \sim N(\mu, \sigma^2)$

$$\pi(\mu, \sigma) \propto \sigma^{-1}$$

Again, Improper Prior; but use of an improper prior can still lead to acceptable posteriors, which are just based on diffuse priors

- Poisson Example

$$Y \mid \lambda \sim \text{Poisson}(\lambda)$$

$$\pi(\lambda) \propto \lambda^{-1}$$

- Most general (**Jeffreys Prior**)

$$\text{Density} : f(y \mid \theta)$$

$$I(\theta) = -E \left(\frac{\partial^2 \log f(y \mid \theta)}{\partial \theta \partial \theta^T} \right)$$

$$\pi(\theta) = \sqrt{\det I(\theta)}$$

Model-Based Justification of the Horvitz-Thompson Estimator

HT – Estimator :

$$q = \sum_i y_i / \pi_i$$

$$y_i \sim (E(y_i) = \beta \pi_i, V(y_i) = \sigma^2 \pi_i^2)$$

q = Weighted Least Squares Estimate
of β (an alternative model-based approach
to estimating a total given a complex sample design)

A General Approach to Bayesian Inference

- Log-likelihood (applies to any model for y):

$f(y | \theta)$ = Probability Density Function (Normal, Poisson, Bernoulli, etc.)

$$l(\theta | y_{inc}) = \sum_i \log f(y_i | \theta)$$

- Maximum Likelihood Estimate (general):

$$\textit{Maximize} : l(\theta | y_{inc})$$

$$\textit{Maxima} : \hat{\theta}_{MLE}$$

- Information Matrix and Variance of Estimates:

$$I(\hat{\theta}) = - \frac{\frac{\partial^2 l(\theta | y_{inc})}{\partial \theta \partial \theta^T}}{\Big|_{\theta = \hat{\theta}}} \quad \hat{V} = I^{-1}(\hat{\theta})$$

- Assume any prior distribution, characterized by:

$$\text{prior mean} = \theta_P$$

$$\text{prior variance} = V_P$$

- The posterior distribution then has these characteristics:

posterior mean :

$$E(\theta | y_{inc}) \square (\hat{V}^{-1} + V_P^{-1})^{-1} (\hat{V}^{-1} \hat{\theta} + V_P^{-1} \theta_P)$$

posterior Variance :

$$V(\theta | y_{inc}) \square (\hat{V}^{-1} + V_P^{-1})^{-1}$$

- Given a large sample, we can assume that the posterior distribution is approximately **normal**:

$$\theta \mid y_{inc} \sim N[E(\theta \mid y_{inc}), V(\theta \mid y_{inc})]$$

- **Simulating Populations:** draw theta from posterior given data, then draw values of Y for excluded cases:

$$\theta \sim N[E(\theta \mid y_{inc}), V(\theta \mid y_{inc})]$$

$$Y_j \mid \theta \sim f(y \mid \theta), j = n + 1, n + 2, \dots, N$$

- When posteriors have a **known** form, point estimation and inference can be based directly on the posterior distribution:

$$\theta = (\mu, \sigma)$$

$$\pi(\theta | y_{inc}) \propto N(\hat{\theta}, \hat{V})$$

$$\hat{\theta} = \text{Mode of } \pi(\theta | y_{inc})$$

$$\hat{V}^{-1} = \text{Second derivative of} \\ \text{the logarithm of } \pi(\theta | y_{inc})$$

- For more **general** cases, with complex priors and likelihoods, hard (if not impossible) to write down the posterior
- Need Iterative simulation techniques such as **Gibbs Sampling** and **Markov Chain Monte Carlo (MCMC)** methods; we'll see examples shortly

Examples of Bayesian Analysis in R

- Fit a multilevel model using the `MCMCglmm` package:

```
library(MCMCglmm)
```

```
model3.3.bayes <- MCMCglmm(weight ~ sex1 + litsize + treatment, random = ~litter, data =  
ratpup, rcov = ~idh(treatment):units)
```

- By default, `MCMCglmm` uses 13,000 MCMC iterations, resulting in several thousand (10,000) simulated draws of the parameters from their posterior distribution based on the specified model and **non-informative** prior distributions. A default of 3,000 burn-in draws are used for the MCMC algorithm (**burn-ins** are used to help stabilize convergence of simulated draws from the posterior; initial draws may be highly unreliable). We can plot the **post** burn-in draws of these parameters using a simple plot function:

```
plot(model3.3.bayes)
```

- Users can hit ENTER to cycle through the plots, which show the sequence of draws for each parameter (which should stabilize over iterations, indicating convergence of the sequence to random draws around an overall mean) and the density function defined by all of the posterior draws of each parameter. The density functions provide an initial sense of whether the distribution of the draws of a given parameter from the posterior distribution for all of the parameters in the model includes 0 as a plausible value.

Examples of Bayesian Analysis in R

- To generate inferences, we can apply the `summary()` function to the object generated by `MCMCglmm()`:

```
summary(model3.3.bayes)

Iterations = 3001:12991
Thinning interval = 10
Sample size = 1000

DIC: 313.9606

G-structure: ~litter

      post.mean l-95% CI u-95% CI eff.samp
litter    0.1066  0.04921  0.1813      884

R-structure: ~idh(treatment):units

      post.mean l-95% CI u-95% CI eff.samp
Control.units    0.2694  0.20757  0.3388    1000
High.units       0.1116  0.07475  0.1586    1000
Low.units        0.0861  0.06453  0.1080    1000

Location effects: weight ~ sex1 + litsize + treatment

      post.mean l-95% CI u-95% CI eff.samp pMCMC
(Intercept)    8.32723  7.78502  8.91543    1000 <0.001 ***
sex1           -0.34294 -0.43046 -0.26315    1000 <0.001 ***
litsize        -0.13026 -0.16666 -0.09339    1000 <0.001 ***
treatmentHigh -0.85954 -1.23652 -0.47429    1000 <0.001 ***
treatmentLow  -0.43554 -0.72088 -0.12940    1000  0.006 **
---
Signif. codes:  0  ***  0.001  **  0.01  *  0.05  .  0.1  1
```

- The DIC criterion can be used to compare competing models, similar to the AIC and BIC criteria. We see means of the 1,000 draws of each parameter from the posterior distribution, along with 95% credible intervals for the parameters in the model.

Examples of Bayesian Analysis in R

- Now, fit a multilevel model using the `brms` package:

```
library(brms)
```

```
fit1 <- brm(vsae ~ age.2 + I(age.2^2) + sicdegp2.f + age.2 * sicdegp2.f +  
I(age.2^2) * sicdegp2.f + (age.2 + I(age.2^2) | childid),  
data = autism.updated, family = gaussian("identity"),  
prior = c(set_prior("normal(0,10)", class = "b"), set_prior("cauchy(0,2)", class =  
"sd"), set_prior("lkj(2)", class = "cor")),  
warmup = 500, iter = 2000, chains = 2)
```

- Note: clearly-specified distribution for the dependent variable, VSAE (Normal, or Gaussian)
- Clearly-specified prior (mostly non-informative) distributions, for the fixed effects [N(0,10)], variance components [Cauchy(0,2)], and correlations of the random effects [LKJ(2)]; **subjective!!! As informative as you want, given your prior knowledge! But these choices set up for a joint posterior that cannot be specified analytically. Use MCMC.**
- **500** burn-in draws, followed by 1,500 additional draws, with **2** concurrent chains of draws to assess convergence of the draws to some stable distribution

Examples of Bayesian Analysis in R

```
summary(fit1, waic=T)
```

```
Family: gaussian
Links: mu = identity; sigma = identity
Formula: vsae ~ age.2 + I(age.2^2) + sicdegp2.f + age.2 * sicdegp2.f + I(age.2^2) * sicdegp2.f + (age.2 + I(age.2^2) | childid)
Data: autism.updated (Number of observations: 610)
Samples: 2 chains, each with iter = 2000; warmup = 500; thin = 1;
total post-warmup samples = 3000
ICs: LOO = NA; WAIC = 4219.42; R2 = NA
```

Group-Level Effects:

~childid (Number of levels: 158)

	Estimate	Est.Error	1-95% CI	u-95% CI	Eff.Sample	Rhat
sd(Intercept)	0.90	0.52	0.05	1.95	172	1.01
sd(age.2)	3.75	0.36	3.07	4.49	835	1.00
sd(Iage.2E2)	0.38	0.04	0.30	0.45	810	1.00
cor(Intercept, age.2)	0.05	0.30	-0.49	0.65	101	1.01
cor(Intercept, Iage.2E2)	0.44	0.35	-0.46	0.88	35	1.03
cor(age.2, Iage.2E2)	-0.31	0.12	-0.52	-0.07	744	1.00

Population-Level Effects:

	Estimate	Est.Error	1-95% CI	u-95% CI	Eff.Sample	Rhat
Intercept	13.74	0.83	12.12	15.41	2025	1.00
age.2	5.59	0.77	4.11	7.04	952	1.00
Iage.2E2	0.20	0.08	0.05	0.36	952	1.00
sicdegp2.f1	-5.36	1.11	-7.53	-3.23	3000	1.00
sicdegp2.f2	-3.99	1.09	-6.19	-1.91	3000	1.00
age.2:sicdegp2.f1	-3.31	1.07	-5.38	-1.24	1067	1.00
age.2:sicdegp2.f2	-2.71	1.02	-4.70	-0.67	1131	1.00
Iage.2E2:sicdegp2.f1	-0.13	0.12	-0.36	0.10	977	1.00
Iage.2E2:sicdegp2.f2	-0.13	0.10	-0.34	0.07	1022	1.00

Family Specific Parameters:

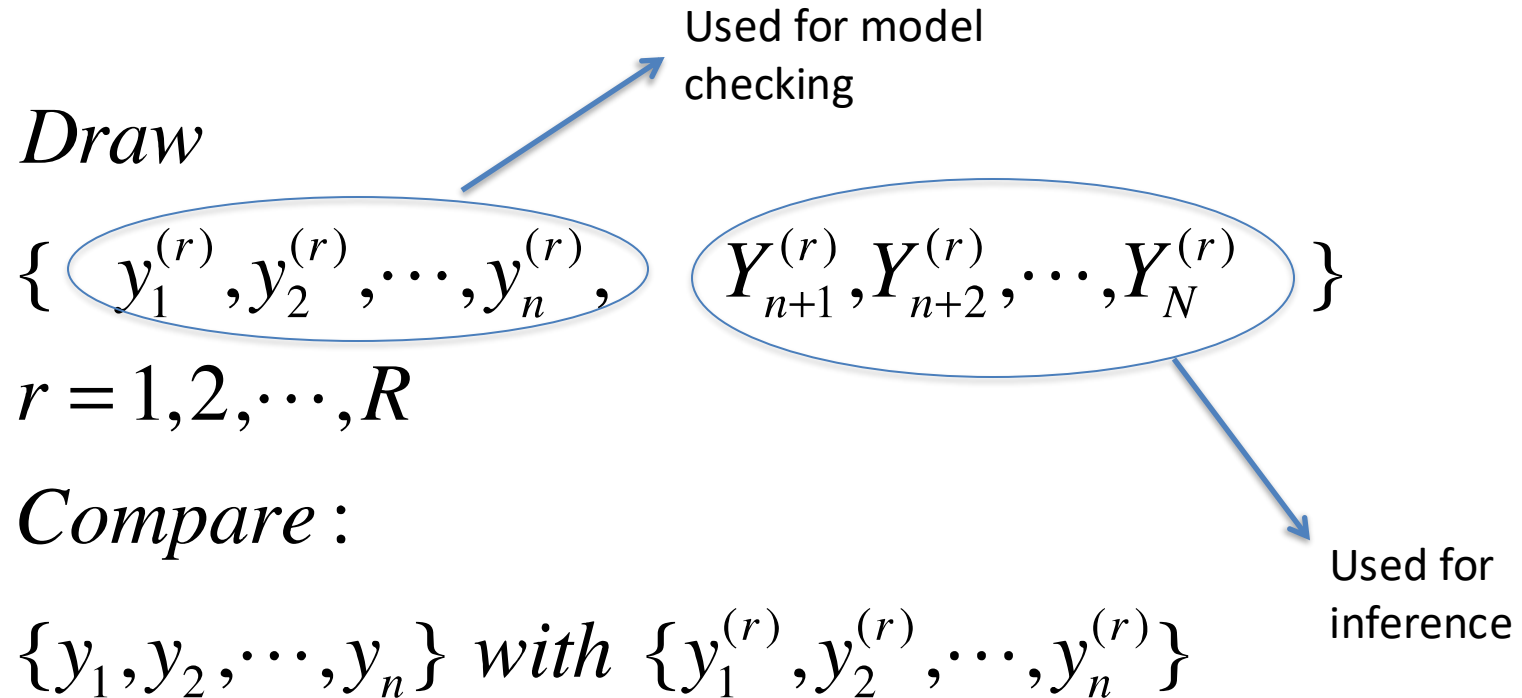
	Estimate	Est.Error	1-95% CI	u-95% CI	Eff.Sample	Rhat
sigma	6.21	0.25	5.76	6.74	1140	1.00

- Note: much more natural inference about variance components and unusual parameters (e.g., correlations of random effects) than is possible in frequentist approaches, which rely on likelihood ratio tests, mixtures of chi-square distributions as null distributions, controversial Wald tests, etc.
- Just simulate draws of the parameters from their posterior distributions, and form 95% credible intervals! The probability that the parameter lies between these values is 0.95.
- **Key question: is model correct?**

Calibrated Bayes (Little, 2006)

- Survey Design is very important
 - Probability Sample Design provides ethically justifiable study of population
 - Stratification improves representation
- Model-Based inference provides the right framework for defining the unknown quantity (source of our uncertainty) and drawing inference about the unknown quantity
- “Good” properties of repeated sampling, such as unbiasedness, minimum variance, design consistency, etc., are desirable

- A balanced view is to develop the best possible design, best possible Bayesian inference, and check using the repeated sampling framework
- Idea:



- Are the observed data consistent with the data from repeated simulations under the specified model?

Calibrated Bayes: Example

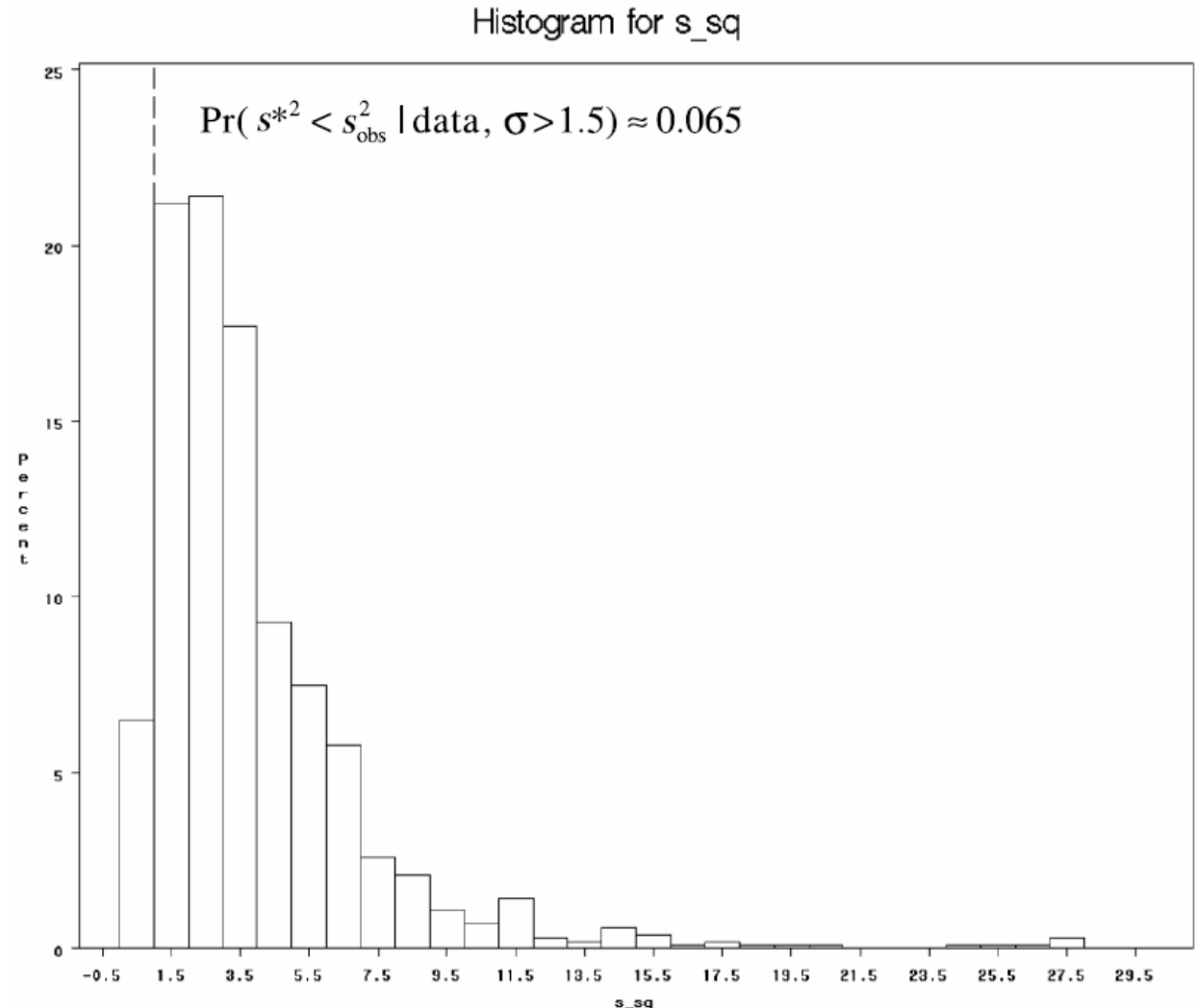
- **Goal:** Are the observed data consistent with repeated simulations from the fitted model? Idea of **posterior predictive checking**
- Example from Little (2006):
 - SRS of i.i.d. Normal values, $n = 7$, sample mean = 1, sample SD = 1
 - Freq. 95% CI using sample estimates: $1 \pm 2.447 \times (1 / \sqrt{7}) = 1 \pm 0.92$
 - Identical Bayes Credible Interval: Jeffreys prior with $p(\text{mean}, \text{SD}) \sim 1/\text{SD}$ (**Slide 7**)
 - What if we are told that SD is known to be 1.5 (not estimated)? No uncertainty in estimation of the sample SD, and 95% CI is $1 \pm 1.96 \times (1.5 / \sqrt{7}) = 1 \pm 1.11$
 - Now, we learn that known SD is something greater than 1.5. We could use one of the intervals above, or a credible interval with the prior updated: 1 ± 1.45

Calibrated Bayes: Example

- Example from Little (2006), cont'd:
 - What do we choose? First interval provides exact 95% frequentist coverage of true mean, assumes that SD is not known, but how can this be narrower than an interval that actually incorporates known information about the SD?
 - Second interval is an asymptotic frequentist approximation, given that the SD is no longer estimated (but our sample is only $n = 7!!!$). Wouldn't coverage of second interval be less than 95% if we know that SD is greater than 1.5?
 - Third (Bayesian credible) interval incorporates the prior constraint and is thus wider, but depends entirely on the prior information being correct.
 - How can we marry these ideas for reasonable inference?

Calibrated Bayes: Example

- Note the **posterior predictive distribution** (simulate repeated samples under the Bayesian model, compute the sample variance for each) for the sample variance under the Normal model for the variable of interest, with a Jeffreys prior
- The posterior probability of observing a sample variance of 1 or lower is about 0.065, which is low but not unusually low, so the Bayesian model used seems reasonable



Superpopulation Inferences

- Consider FinitePopulation as a sample from a superpopulation (Model is interpreted as this “first stage” sampling of a finite population)
- Designed Sample as the “second stage” sampling from the realized finite population
- Use both stages of sampling to evaluate the properties of the procedures
- Apply the procedure to the data in hand

Superpopulation Inferences

- Two expectations:
 - One with respect to the repeated sampling distribution over the indicator I (second stage)
 - One with respect to the repeated draws of the finite population from the superpopulation (repeated sampling with respect to model M)

$$E(q) = E_M(E_I(q))$$

Superpopulation Variance

- Variance is also computed using two stages of sampling:

$$Var(q) = E_M(V_I(q)) + V_M(E_I(q))$$

$$\text{Usually, } V_M(E_I(q)) \approx O(1/N),$$

can be ignored

- Example: $Y_i \mid \mu, \sigma^2 \sim iid \ (\mu = E(Y), \sigma^2 = Var(Y))$

Simple random sample

$$Var(\bar{y}) = (1 - f)E_M(S^2) / n + \sigma^2 / N$$

$$S^2 = \sum_{j=1}^N (Y_j - \bar{Y})^2 / (N - 1)$$

Under normality :

$$E_M(S^2) = \sigma^2$$

$$Var(\bar{y}) = \sigma^2 / n$$

$$Est.Var(\bar{y}) = s^2 / n$$

$$s^2 = \frac{1}{n-1} \sum_i (y_i - \bar{y})^2$$

Confidence Interval

$$\bar{y} \pm z_{\alpha/2} \sqrt{s^2 / n}$$

Repeated sampling interpretation involves both stages or phases of sampling